

Math 241, F12
Quiz 6

Name: Answer
SSN: xxx-xx-

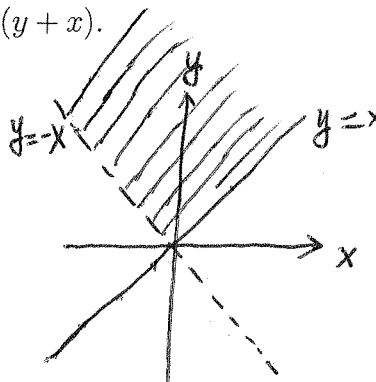
Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

Problem 1. (4 points) Find and sketch the domain of the function

$$f(x, y) = \sqrt{y-x} \ln(y+x).$$

$$\begin{cases} y-x \geq 0 \Rightarrow y \geq x \\ y+x > 0 \Rightarrow y > -x \end{cases}$$

$$D = \{(x, y) \mid y \geq x \text{ and } y > -x\}$$



Problem 2. (6 points) Find the limit if it exists or show that the limit does not exist.

$$(a) \lim_{(x,y) \rightarrow (1,-1)} e^{-xy} \cos(x+y)$$

$$= e^{-1 \cdot (-1)} \cos(1+(-1))$$

$$= e \cos(0)$$

$$= e$$

$$(b) \lim_{(x,y) \rightarrow (0,0)} \frac{y^4}{x^4 + 3y^4}$$

when $(x, y) \rightarrow (0, 0)$ along the x -axis,

$$\frac{y^4}{x^4 + 3y^4} = \frac{0^4}{0^4 + 3(0^4)} = 0 \rightarrow 0$$

when $(x, y) \rightarrow (0, 0)$ along the y -axis

$$\frac{y^4}{x^4 + 3y^4} = \frac{y^4}{0^4 + 3y^4} = \frac{1}{3} \rightarrow \frac{1}{3}$$

Thus $\lim_{(x,y) \rightarrow (0,0)} \frac{y^4}{x^4 + 3y^4}$ does not exist.